

# A Machine Learning Approach to Twitter Sentiment Analysis

## Data Preparation, Cleaning, and Feature Generation Report

CSCE 3602

Kirolous Fouty  
Computer Engineering  
Department  
American University in Cairo  
Cairo, Egypt  
kirolous\_fouty@aucegypt.edu

Mohamed Sabry  
Computer Engineering  
Department  
American University in Cairo  
Cairo, Egypt  
momo12320@aucegypt.edu

### 1 Full description of your dataset of choice and the rationale for choosing it

#### 1.1 Sentiment140 Dataset Description

Sentiment140 is a widely used dataset for sentiment analysis on Twitter data. It consists of 1.6 million labeled tweets and offers valuable insights for understanding public sentiment on various topics. The dataset was created by classifying the tweets as positive, negative, or neutral based on the presence of emoticons. Each tweet is labeled as follows:

- 0 for negative sentiment
- 2 for neutral sentiment
- 4 for positive sentiment

The dataset includes various fields:

- Tweet ID: A unique identifier for each tweet
- Sentiment: The polarity label (0, 2, or 4)
- Date: The date when the tweet was posted
- Query: The search term used (if applicable)
- User: The Twitter username
- Text: The actual content of the tweet

#### 1.2 Rationale for Choosing Sentiment140

1. Large-Scale Data: With 1.6 million labeled tweets, Sentiment140 provides a substantial dataset for training machine learning models, which helps improve the performance and generalizability of sentiment analysis models.

2. Real-World Data: The dataset consists of real tweets, making it ideal for projects focused on natural language processing tasks that require understanding how people express opinions on social media.
3. Balanced Sentiment Classes: Sentiment140 includes positive, negative, and neutral labels, allowing for a comprehensive analysis of sentiment across different spectrums.
4. Pre-Labeled Data: Since the dataset is already labeled, it saves time and effort on manual annotation, enabling a quicker start to model training and experimentation.
5. Widely Recognized: Sentiment140 is a well-known and widely cited dataset in the field of sentiment analysis, making it a reliable choice for benchmarking and comparing model performance.

These characteristics make Sentiment140 an excellent choice for building, training, and evaluating sentiment analysis models on Twitter data.

### 2 Full analysis of the different features, including but not limited to: statistical distribution, missing values, correlation with label, unique values for nominal features, semantic importance / relevance, etc.

#### 1. Tweet ID

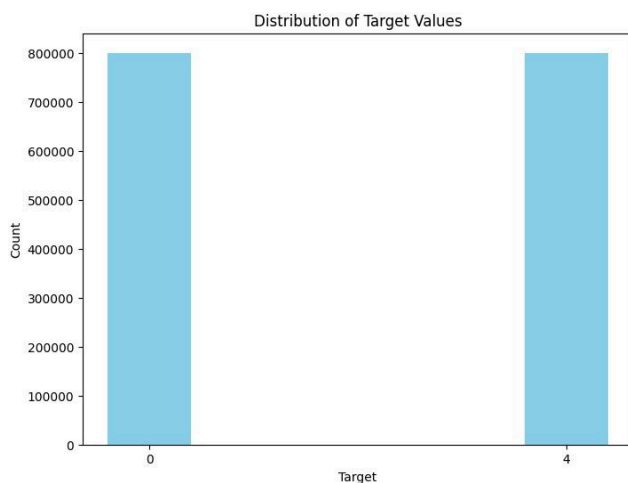
- Type: Numerical (Identifier)
- Description: A unique identifier for each tweet.
- Statistical Distribution: No meaningful distribution as it's just a unique identifier.
- Missing Values: None, as each tweet has a unique ID.

- Correlation with Label: None, since it doesn't provide any semantic information about the sentiment.
- Unique Values: All values are unique.
- Semantic Importance: None; this feature is typically ignored in model training since it doesn't contribute to sentiment analysis.

## 2. Sentiment (Label)

- Type: Categorical (Nominal)
- Description: Indicates the sentiment of the tweet, with possible values: 0 (negative), 2 (neutral), and 4 (positive).
- Statistical Distribution: Typically balanced in the original dataset with approximately equal representation of the positive and negative classes:
  - Negative (0): 800k unique entries ~50%
  - Neutral (2): 0 entries ~0%
  - Positive (4): 798k unique entries ~50%
  - It's important to validate this distribution to ensure there's no imbalance.
- Missing Values: None, as it's the target label.
- Unique Values: Two unique values (0, 4).
- Semantic Importance: High; this is the target variable for the sentiment analysis task.

The following graph provides insight on the distribution of the target sentiment (labels).



## 3. Date

- Type: Datetime
- Description: The date and time when the tweet was posted.
- Statistical Distribution: Dates can be examined to determine temporal patterns, such as peaks in positive or negative sentiment over time.
- Missing Values: None.

- Correlation with Label: Potential correlation if sentiment shifts during specific time periods (e.g., events causing spikes in positive or negative tweets).
- Unique Values: A mix of unique and repeated values, depending on tweet activity during certain periods.
- Semantic Importance: Low to medium; could be useful for time-based sentiment trends but is generally not a strong predictor of individual tweet sentiment.

## 4. Query

- Type: Categorical (Nominal)
- Description: The search term used to collect the tweet.
- Statistical Distribution: Generally not diverse, as it always contains the value "NO\_QUERY" (indicating no specific search term was used).
- Missing Values: None.
- Correlation with Label: Minimal; since all entries are "NO\_QUERY," it doesn't provide useful insight into sentiment.
- Unique Values: only "NO\_QUERY."
- Semantic Importance: Low; it doesn't contribute much to sentiment prediction.

## 5. User

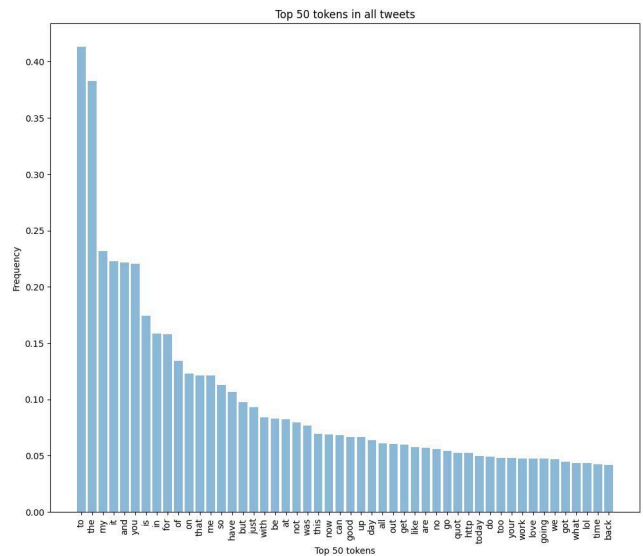
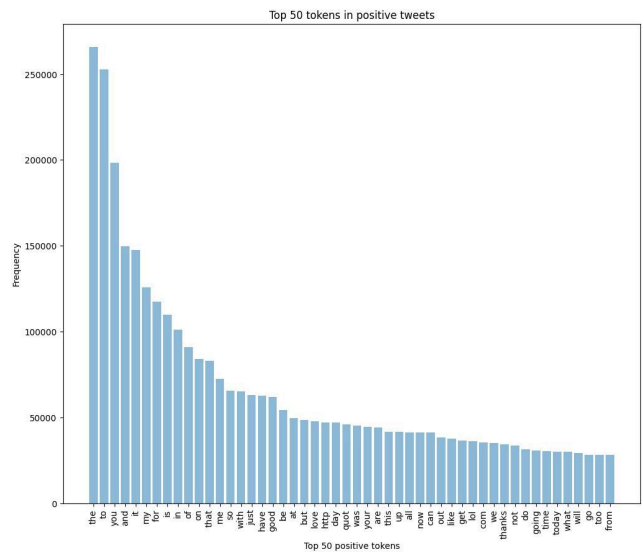
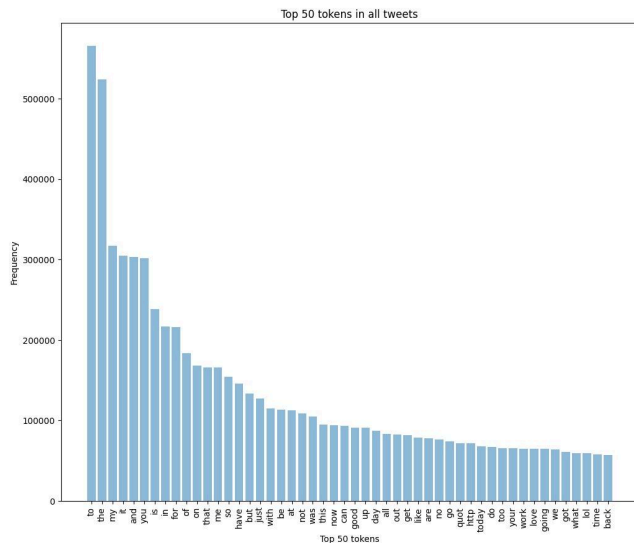
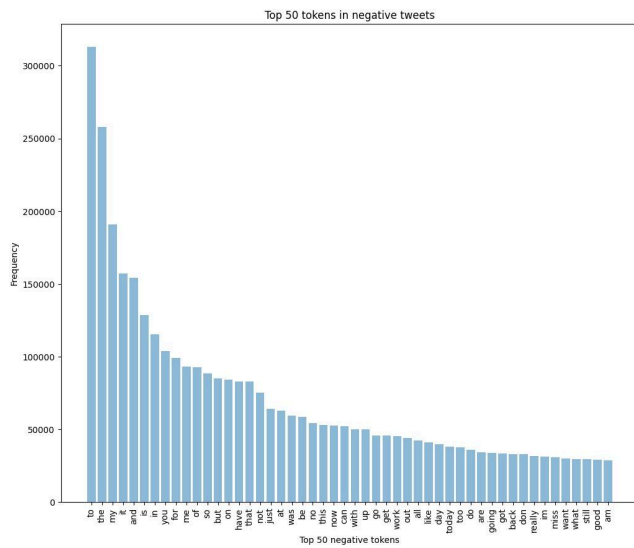
- Type: Text data
- Description: The username of the person who posted the tweet.
- Statistical Distribution: skewed distribution with many unique users, but some posting more frequently.
- Missing Values: None.
- Correlation with Label: Can be explored to see if certain users consistently post more positive or negative tweets, but overall, the correlation is weak.
- Unique Values: about 660k.
- Semantic Importance: Low to medium; while it might reveal patterns of sentiment for specific users, it's usually not directly relevant for general sentiment classification.

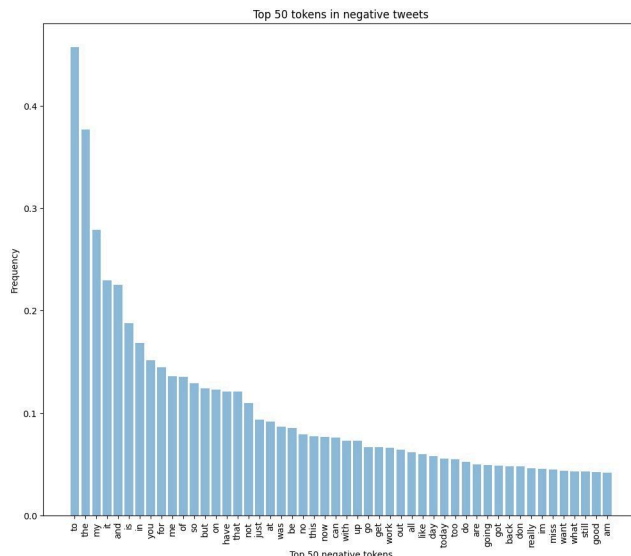
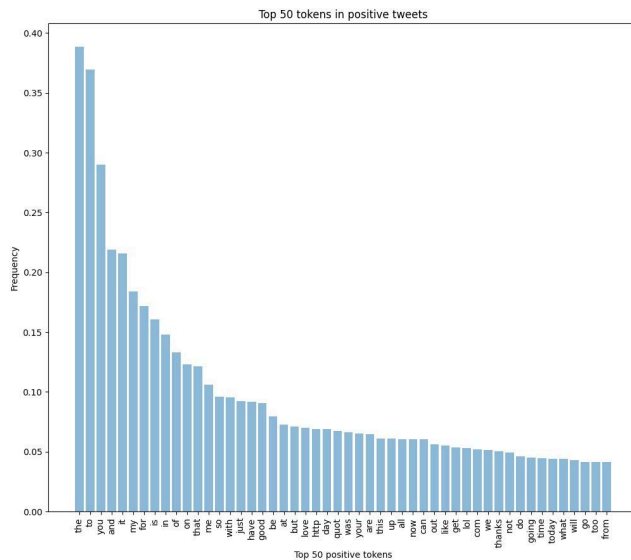
## 6. Text

- Type: Text data
- Description: The actual content of the tweet.
- Statistical Distribution: The length of tweets ranges from 1 to 140 characters (given Twitter's character limit during the dataset's collection period).
- Missing Values: a little few, about 270 tweets.
- Correlation with Label: Highly correlated; words, phrases, and emoticons in this feature are strong indicators of sentiment.

- Unique Values: Nearly every tweet is unique, but common phrases, words, and hashtags will appear multiple times.
- Semantic Importance: Extremely high; this is the primary feature for sentiment analysis, as it contains the linguistic elements that convey emotion, opinion, or sentiment.

The following graphs provide insight on the dataset.





## Text Analysis Details

- **Word Frequency:** Analyze word frequency distributions for positive, neutral, and negative labels. Common positive words (e.g., "happy," "love"), negative words (e.g., "sad," "hate"), and neutral terms will surface.
- **N-grams:** Analyze unigrams, bigrams, and trigrams to capture phrases indicative of different sentiments.
- **Emoticons:** Check for emoticon frequency and their correlation with sentiment labels (e.g., ":") in positive tweets).

- **Sentiment Score:** Use pre-trained sentiment lexicons to assign sentiment scores to words and validate correlations with the labels.

## Feature Engineering Recommendations

- Convert the Text into numerical features using techniques such as TF-IDF, Bag-of-Words, or word embeddings (Word2Vec, GloVe, or BERT).
- Extract additional features like the length of the tweet, the number of hashtags, and the presence of certain keywords or emoticons.

## Missing Values Overview

- A little few tweets are empty, about 270 tweets. We will drop these entries.

## Correlation Analysis

- The primary feature correlating with the sentiment label is the Text itself, as the other features (e.g., date, user) have little direct correlation.

## Unique Values Summary

- Tweet ID: All unique
- Sentiment: Two unique values (0, 4)
- Date: Various unique values
- Query: Always "NO\_QUERY"
- User: A high number of unique values
- Text: Nearly all entries are unique

## Semantic Importance and Relevance

- The most semantically relevant feature for sentiment analysis is the Text feature, as it directly contains the expressions and opinions that the model aims to interpret.

The Sentiment140 dataset's main predictive power lies in the textual data, which is highly indicative of sentiment. While the other features provide context, they contribute less to the classification task.

The dataset's balanced sentiment positive and negative classes and complete data make it ideal for training robust sentiment analysis models.

## 3 Full description of the cleaning and pre-processing steps done to prepare the dataset

### 1. Data Loading

Importing the Dataset: Load the dataset into a DataFrame

using libraries such as Pandas, ensuring all features (Tweet ID, Sentiment, Date, Query, User, and Text) are properly imported.

## 2. Handling Missing Values

Check for Missing Data: Verify if there are any missing values in each column. Although Sentiment140 typically has no missing values, this step confirms the dataset's integrity.

## 3. Dropping Irrelevant Columns

Irrelevant Columns: Drop columns that don't provide valuable information for sentiment analysis:

- Tweet ID: Unique identifier with no predictive power.
- Date: Provides temporal information but is not directly relevant for sentiment classification.
- Query: Always contains "NO\_QUERY" and doesn't contribute to the analysis.
- User: Twitter username isn't meaningful for sentiment prediction.

Result: Retain only the Target Sentiment and Text columns.

## 4. Label Encoding

Converting Sentiment Values: Transform the sentiment labels (0, 2, 4) into more manageable categories:

Convert 4 to 1 (Positive)

Convert 0 to 0 (Negative)

Drop 2 (Neutral) based on our desire of a binary classification (positive vs. negative).

## 5. Text Cleaning

The Text feature contains raw tweets, which include noise such as URLs, mentions, hashtags, punctuation, and special characters. Our approach to data cleaning:

- Lowercasing: Convert all text to lowercase to maintain uniformity (e.g., "Happy" and "happy" are treated the same).
- Removing URLs: Use regular expressions (e.g., `r"http\S+|www\S+"`) to identify and remove URLs.
- Removing Mentions: Remove Twitter handles (e.g., `@username`) using regex (`r"@w+"`), as they don't convey sentiment.
- Removing stop words: Remove those that are of little to no sentimental value.
- Removing emojis: To be deleted or replaced with text.
- Removing Hashtags: Strip the '#' symbol from hashtags but keep the actual words (e.g., `#happy` → `happy`) since they might convey sentiment.
- Removing Non-alphabetic Characters: Remove characters that might confuse the model.

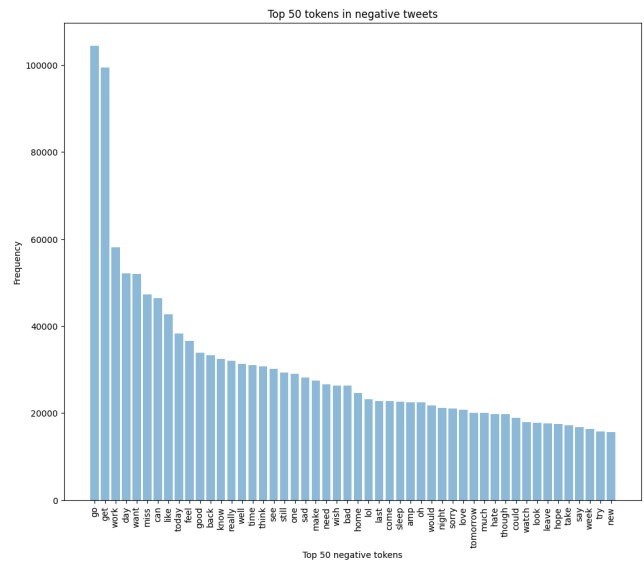
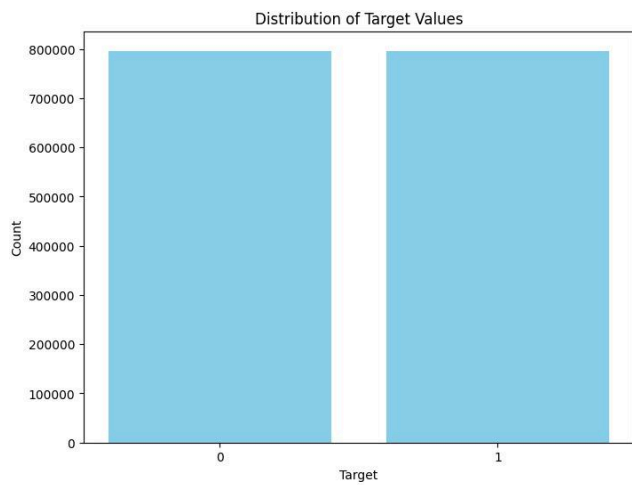
- Removing Repeated Letters: Removing accidentally or intentionally repeated letters in desire of generalization.
- Removing Grammatical Contraction: Expanding contracted forms like "don't" to "do not" in text.
- Lemmatization/Stemming: Reducing words to their root form to capture the core meaning, e.g., "running" becomes "run".
- Tokenization: Dividing the cleaned text into individual words (tokens) using tokenization techniques from libraries like NLTK or spaCy. This process prepares the text for vectorization.
- Saving the Cleaned Data: Save the cleaned and preprocessed dataset to disk for efficient reuse.

These steps will help transform raw Twitter data from the Sentiment140 dataset into a structured format suitable for building and training sentiment analysis models.

## 4 Final list of chosen features, the rationale for adding or removing any feature, and a description of the dataset size and metrics post-cleaning

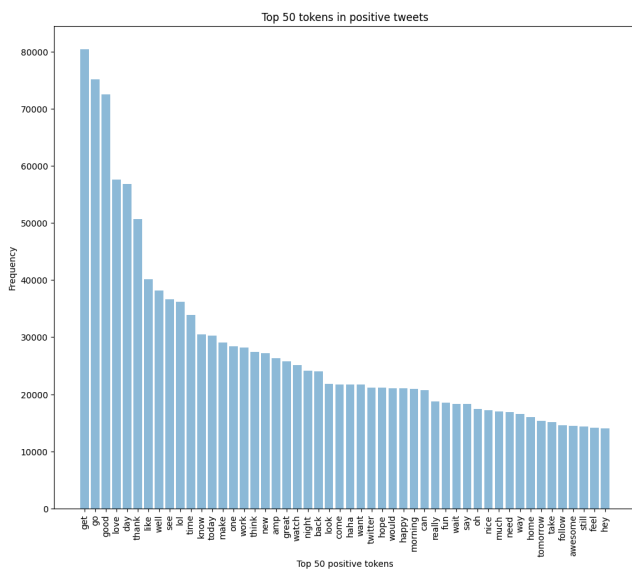
For the final feature set, we dropped the original dataset's Tweet ID, Query, Date, and User columns. Since we are working with natural language processing, these other features do not offer any value to us as the IDs can be replaced with the dataset indexing. In contrast, the other entries are only related to the tweet itself and not its sentimentality. After preprocessing but before vectorization, we lost only 7181 rows that were made up of tweets that contained nothing more than mentions. This made it so that the number of entries for negative tweets became 796450 and the number of positive tweets became 796369. We have visualized the target value distribution in the following graph.

Distribution of Target Values according to the Target:

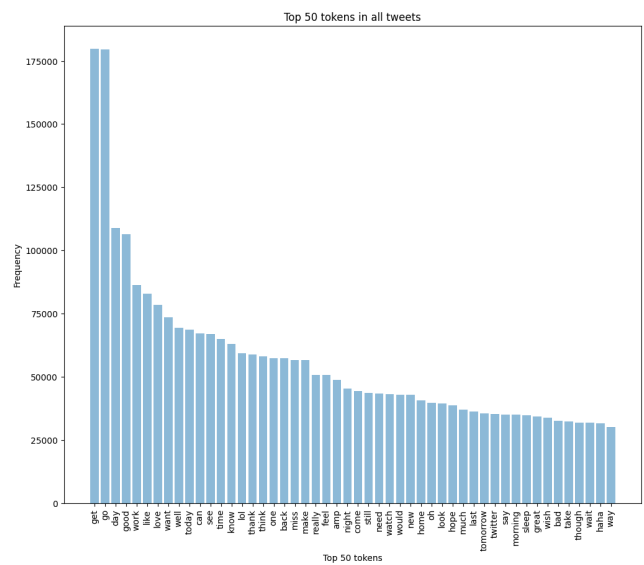


After vectorization, the final form of the data is only 242934 rows. We decided on vectorizing for TFIDF and unweighted vectorization so that we may use TFIDF for the next milestone and other models. You can see the frequency of appearance of the top 50 tokens from each category of a tweet as well as the token frequency for all of the tweets.

Frequency of top 50 most frequent tokens in positive tweets:

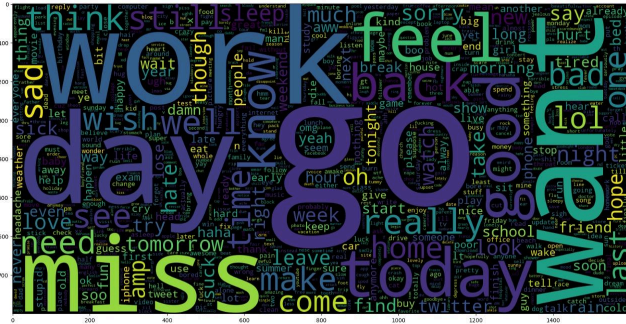


Frequency of top 50 most frequent tokens for all tweets:



Frequency of top 50 most frequent tokens in negative tweets:

Word cloud for the top 1000 tokens in negative tweets:



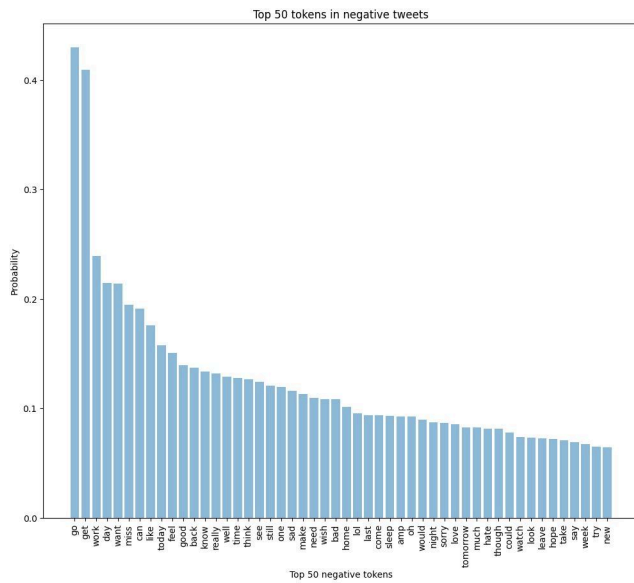
Top 50 tokens in positive tweets

| Token   | Probability |
|---------|-------------|
| get     | 0.33        |
| go      | 0.31        |
| good    | 0.30        |
| love    | 0.24        |
| day     | 0.23        |
| thank   | 0.21        |
| best    | 0.16        |
| well    | 0.15        |
| see     | 0.14        |
| to      | 0.13        |
| time    | 0.12        |
| know    | 0.12        |
| best    | 0.11        |
| make    | 0.11        |
| one     | 0.11        |
| think   | 0.10        |
| new     | 0.10        |
| great   | 0.09        |
| watch   | 0.09        |
| back    | 0.08        |
| book    | 0.08        |
| love    | 0.08        |
| have    | 0.08        |
| want    | 0.08        |
| best    | 0.08        |
| love    | 0.08        |
| would   | 0.08        |
| happy   | 0.08        |
| more    | 0.08        |
| can     | 0.08        |
| really  | 0.07        |
| wait    | 0.07        |
| day     | 0.07        |
| or      | 0.07        |
| nice    | 0.07        |
| much    | 0.07        |
| way     | 0.07        |
| home    | 0.06        |
| take    | 0.06        |
| follow  | 0.06        |
| awesome | 0.06        |
| still   | 0.06        |
| feel    | 0.06        |
| hey     | 0.06        |

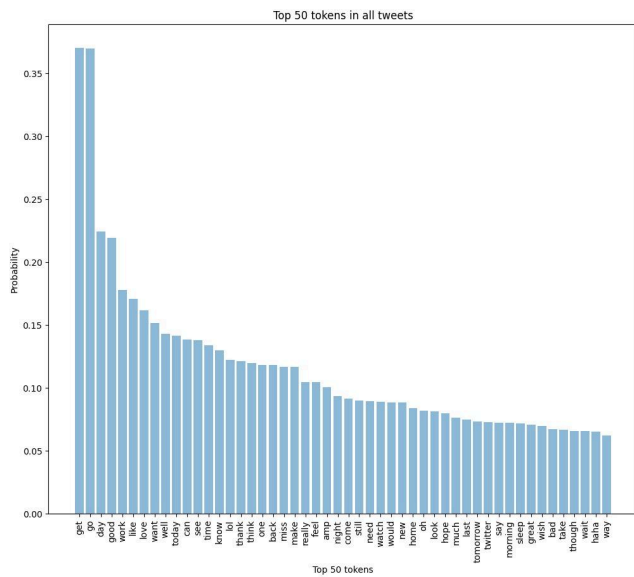
Top 50 positive tokens

Probability of top 50 most frequent tokens appearing in negative tweets:

We also checked if some words showed up in more than 85% of the entries, but we could not find any that fit this criteria in order to remove them. Here are the probability graphs for the top 50 tokens attached below for each of the categories and the entire dataset.



Probability of top 50 most frequent tokens appearing in all tweets:



We tried to calculate the correlation between each word and the target label, however, it took more than 700 minutes and did not finish, so we needed more time to completely calculate the correlations and store them in a list. The final dataset that the model will take will be one of the 2 .pkl files that contain the vectorizer, as well as a CSV file with the labels.